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Paleomagnetism of the Devonian Kennett Limestone and the rotation of the eastern Klamath Mountains, California

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The Devonian Kennett Limestone in the Eastern Klamath terrane in northern California has a stable magnetization that was acquired after the folding of the formation apparently in the very early Cretaceous. The mean paleomagnetic vector has the direction $I = -49°$, $D = 219°$, $A_{95} = 7.2°$ and the parameters of the corresponding paleomagnetic pole are latitude = 57°, longitude = 332°, $A_{95} = 7.4°$. All of the stably magnetized specimens have a single polarity. We propose that the magnetization be associated with the emplacement of the Shasta Bally batholith during the M17 reversed interval at 134 Ma. The firmest conclusion that can be drawn from these data is that the Eastern Klamath terrane has undergone $70 \pm 11°$ of clockwise rotation, since the Early Cretaceous. This result does not support earlier tectonic models in which either the entire Klamaths or terranes on the south side of the Klamaths rotated in a counterclockwise sense. The close agreement between the amount of clockwise rotation and that observed in the Oregon Coast Range to the north of the Klamaths suggests that both may reflect the same tectonic process. The data may suggest a few hundred kilometers of northwestward translation of the terrane since the Early Cretaceous.

1. Introduction

The concept that much of western North America is a mosaic of stratigraphically and tectonically distinct terranes is currently providing much insight into the tectonic and geologic history of western North America. More than 50 such terranes have been identified, each characterized by internal homogeneity and continuity of stratigraphy, tectonic style, and history [1]. Paleomagnetic research has shown that many of these terranes have undergone rotations about local vertical axes and that a few of them are clearly allochthonous, having undergone substantial northward displacements [2]. Coney et al. [1] describe as "suspect terranes" a large group of terranes comprising 70% of the Cordillera which are either known to be or are suspected of being allochthonous. One of these suspect terranes is the Eastern Klamath terrane seen as the stippled pattern (*KL*) in Fig. 1 surrounded by other suspect terranes [1]. We have undertaken a paleomagnetic study of the Devonian Kennett Limestone in the Eastern Klamath terrane to determine whether it has undergone a measurable rotation or translation.

2. Regional tectonics

The Eastern Klamath terrane is the most easterly of a set of arcuate terranes separated by major eastward dipping thrust faults. Both the terranes and the faults get younger to the west although older faults may have been reactivated during

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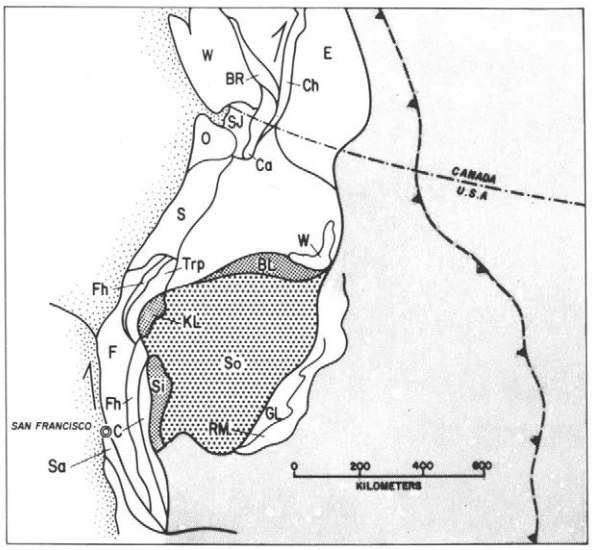


Fig. 1. Generalized map of suspect terranes after Coney et al. [1]. *Shaded area* (unlabeled): autochthonous cratonic basement. *Barbed line*: eastern limit of Mesozoic-Cenozoic deformation. *Coarse stipples*: Sonomia terrane (*So*), of which the Eastern Klamath terrane (*KL*) may be a part [7]. *Fine stipples*: Paleozoic terranes including the Blue Mountains (*BL*) and northern Sierra Nevada (*Si*) thought by Hamilton [5] to once have been joined as part of a north-south-trending linear belt adjacent to an ancestral Sierra Nevada magmatic arc. The names of other terranes shown are as follows. *W* = Wrangellia; *BR* = Bridge River; *E* = Eastern Assemblage; *Ch* = Cache Creek; *SJ* = San Juan; *O* = Olympic; *Ca* = northern Cascades; *S* = Oregon Coast Range; *Trp* = Klamath Mountains (composite); *Fh* = Foothills; *F* = Franciscan; *GL* = Golconda; *RM* = Roberts Mountains; *C* = Calaveras; *Sa* = Salinia.

younger orogenic events. In order of decreasing age the terranes shown in Fig. 1 are the Eastern Klamaths (*KL*), the Central Metamorphic and Western Klamath belts (shown as the composite terrane *Trp*), the Foothill belt (*Fh*), and the Franciscan assemblage (*F*). The Eastern Klamath terrane is thrust over the Devonian rocks of the Central Metamorphic terrane along a thrust fault which has been given a Devonian age [3] and suggested to be associated with the metamorphism of the underlying rocks [4]. But, we observe in Fig. 2 that this fault seems to be cutting the main structural direction (see below) of the Eastern Klamath belt, therefore suggesting that it could have been still active during the Jurassic. The major fault along which the Western Klamath belt is thrust over the Foothill belt and the Franciscan assemblage is early Cretaceous in age. The tectonic history of these belts has been one of crustal shortening by movement along thrust faults for a long period of time.

The contact between the Eastern Klamath terrane and older rocks to the east is hidden beneath a thick cover of volcanic and sedimentary rocks of Cretaceous and Tertiary age. In the absence of direct geologic information about the accretionary history of this terrane, speculations about its early tectonic history have been based mainly on stratigraphic correlations with similar terranes elsewhere in the Cordillera which together with the Eastern Klamath terrane form a larger tectonic pattern. For example, similarities in the faunas and lithologies of middle to late Paleozoic rocks in the Eastern Klamath terrane, the Blue Mountains in eastern Oregon (*BL* in Fig. 1), and the northern Sierra Nevada Mountains (*Si*) have been noted by Hamilton [5], who proposed that these terranes were once part of a continuous belt of Paleozoic rocks along the west flank of an ancestral Sierra Nevada magmatic arc. The Eastern Klamath and Blue Mountain terranes were seen as rifting away from the Sierra Nevada and moving to their present positions in the mid-Tertiary. However, major rifting during the Tertiary as proposed by this model is difficult to reconcile with the presence of late Cretaceous marine sediments between the Sierra Nevada and the rifted Eastern Klamath terrane. The amounts of displacement in this model are too small to be detectable paleomagnetically. However, Hamilton's model predicts clockwise rotation of the Blue Mountains and either no rotation or a small counterclockwise rotation for the Klamath Mountains, both of which are testable with paleomagnetic data.

Concerning the important question of whether the Eastern Klamath terrane originally formed near the continental margin or at a distant locality, opinion has been divided. Churkin [6] considered these rocks to be part of a belt of volcanic and sedimentary rocks extending from Alaska to California and Nevada that formed in a series of marginal ocean basins which opened and closed behind volcanic island arcs from the Cambrian

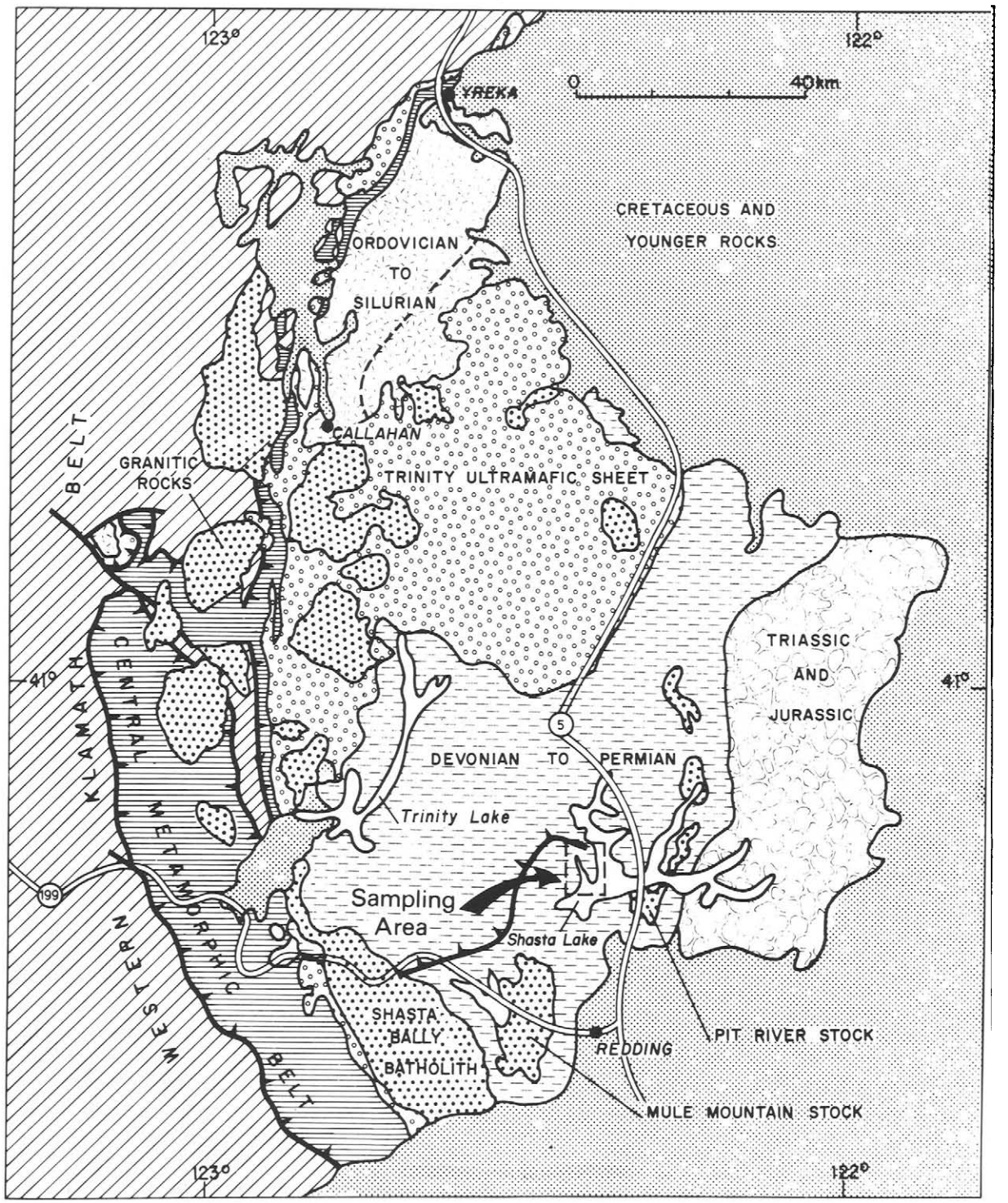


Fig. 2. Generalized geology of the Central Metamorphic belt and the Eastern Klamath terrane. The latter's structure is a southwest-northeast-trending arch with the Trinity Ultramafic Sheet at its center, an Ordovician to Silurian section on its northwest flank, and a Devonian to Jurassic section on its southeast flank (after Irwin, [4]).

through the Devonian. Crustal shortening accompanied by thrusting closed the basins during the Mississippian, after which rifting produced a new series of basins at the end of the Paleozoic. This model predicts no paleomagnetically measurable displacements but allows possible rotations of terranes.

On the other hand Speed [7] proposed that parts of Nevada, California, and Oregon are underlain by a Paleozoic microplate called Sonomia (shown as *So* in Fig. 1) which migrated from a distant site and collided with North America in the early Triassic. The main evidence cited in support of this terrane are the outcrops of Paleozoic rocks in the Eastern Klamath belt and northern Sierra Nevada, some small outcrops of Paleozoic rocks scattered throughout Nevada, and the presence on the eastern margin of the Sonomia terrane of the Golconda terrane, an accretionary wedge that was thrust onto the continental platform in an east-southeast direction in early Triassic time. Speed interprets this as marking the convergence of Sonomia and North America, the polarity of the subduction having the subducting slab dipping to the west. The paucity of Sonomia outcrops Speed attributes to fragmentation and dispersal of the original terrane and burial beneath younger deposits. Similarities in the faunal assemblages and lithologies of many of the scattered outcrops of Paleozoic rocks of Sonomia have suggested to several authors [8,9] that these outcrops were not widely separated when the rocks formed, which would be consistent with the outcrops being either all autochthonous or all allochthonous and part of the same terrane, as Speed proposes. Davis et al. [10] have further noted North American affinities in the upper Paleozoic faunas which they interpret as supportive of the formation of this arc complex in the vicinity of the continent rather than at a transoceanic site. On the other hand, the asynchroneity of volcanic activity in the Eastern Klamath and northern Sierra Nevada terranes during most of the Paleozoic [11,12] suggests that these two terranes were independent island arcs at least until Permian time, when the similarity of the limestones begins to indicate a similar geologic history. Furthermore, isotopic data from Mesozoic plutons suggest an island arc-type lithosphere below Sonomia in Nevada and the northern Sierra Nevada terrane. In contrast, both structural and isotopic data indicate that the Eastern Klamath terrane has behaved, since the Jurassic, like a nappe decoupled from the underlying oceanic lithosphere [7].

The principal structural element of the Eastern Klamath terrane is a southwest-trending arch, the central part of which is formed by the Trinity ultramafic sheet (Fig. 2). Two stratigraphically distinct assemblages of rocks are found on the two flanks of the arch, those to the northwest being early Paleozoic in age and those to the southeast ranging in age from Devonian through Jurassic. Although the ultramafic sheet has been mapped as a single body, the presence of different assemblages of rocks on the two flanks of the arch suggests that the Eastern Klamath terrane may have been built of two different subterranes each bottoming in an ophiolite. The regional structure, although complex, appears to be controlled by southwest-northeast structural trends, suggesting that the direction of convergence during accretion of the belt was toward the southeast. The same convergence direction is indicated by the small thrust fault shown in Fig. 2 west of the sampling area. (For a review of the regional geology, see Irwin [4].)

Our sampling was done on the southeast flank of the arch, where a sequence of Devonian through Jurassic rocks known as the Redding section includes (in stratigraphic sequence) the following lithologies suggestive of an island arc environment: greenstone, rhyolite, shale, limestone, greywacke, andesitic pyroclastics, shale, more limestone, mudstone, tuff, and andesite. Our sampling was in the Kennett Formation, the lowest of the limestone units, which consists of black siliceous shale, rhyolitic tuffs, and limestone. The thickness of the formation is no greater than 120 m although locally it appears thicker due to repetition by faulting. The rocks sampled (Fig. 3) were very fine-grained light to bluish gray limestone which forms the upper part of the formation and is mainly a coral reef with abundant middle Devonian fossils including brachiopods.

Deposition of the Redding section appears to have been nearly continuous except for a probable

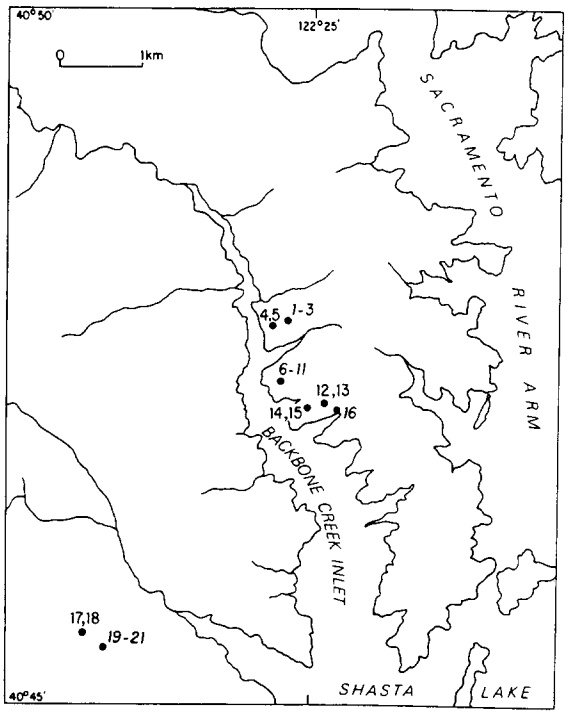


Fig. 3. Sampling sites in the Kennett Formation.

unconformity between the McCloud Limestone and the overlying Permian Nosoni Formation of Mudstone and tuff with minor pyroclastic beds (Fig. 4). A large dike of diorite intrudes the McCloud reducing the outcrops to large blocks embedded in the diorite, therefore hiding the contact of the McCloud with the overlying Nosoni Formation and, in many places, with the underlying Baird Formation. The Dekkas Andesite, which overlies the Nosoni is late Permian to Early Triassic in age and underlies Triassic and Jurassic deposits which comprise the youngest rocks of the Eastern Klamath belt and pass disconformably beneath the Cretaceous sediments of the Great Valley sequence.

Additional igneous activity important in the present context is that of the two younger intrusives described in Table 1, all reasonably close to the sampling area. The closer is the Pit River stock, which intrudes and therefore is younger than the McCloud Limestone, the age of which is no older than Pennsylvanian. This stratigraphic relationship is consistent with the radiometric

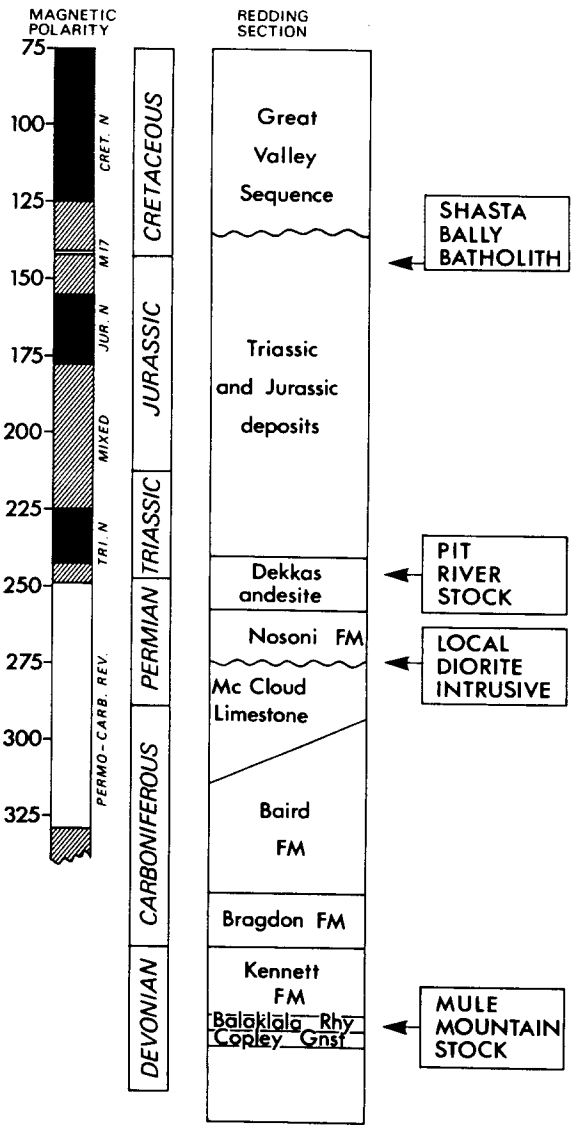


Fig. 4. Generalized stratigraphy of Redding section (after Irwin [4]). Magnetic polarities are shown as follows: *black* = normal polarity; *white* = reversed polarity; *diagonal shading* = mixed polarity. The magnetic polarity time scale is based on Van Hinte [26] from 75 to 156 Ma and on Irving and Pullaiah [36] from 156 to 340 Ma. Both scales were linearly stretched to move the period boundaries to the following ages in Ma, following Armstrong [37]: Cenozoic–65–Cretaceous–143–Jurassic–212–Triassic–247–Permian–289–Carboniferous–367–Devonian–416–Silurian. After stretching to fit these boundaries the ages of the magnetic polarity intervals are (in Ma): 72–Cretaceous normal–126–mixed polarity (M anomalies)–156–Jurassic normal–177–mixed polarity–225–Triassic normal–242–mixed polarity–249–Permo-Carboniferous reversed–329–mixed polarity.

TABLE 1

Intrusives close to sampling area

Distance from sampling area (km)	Name of intrusive	Age (Ma)	Reference
23	Shasta Bally batholith	136	[14]
7	Pit River stock	246	[13]
11	Mule Mountain stock	392	

minimum age of 246 Ma [13]. The Shasta Bally batholith is given an age of 140 Ma on both radiometric and geologic evidence [14] and is clearly associated with the Nevadan orogeny.

3. Paleomagnetism

3.1. Sampling and measurement

A total of 90 oriented cores were collected from the Kennett Limestone at 21 different sites distributed over a distance of 5 km (Fig. 3). Two specimens were cut from 29 of the longer cores, an A specimen at the bottom of the core and a B specimen close to the original surface of the outcrop. Magnetizations were measured on a cryogenic magnetometer. Intensities of natural remanent magnetization (NRM) spanned three orders of magnitude from 10^{-3} to 10^{-5} A/m. Between-site variations in intensity were much larger than within-site variations. Some unusually high intensities due possibly to lightning strikes were found in scattered specimens. Most specimens with either extremely strong or extremely weak intensities of NRM tended to have directions roughly parallel to the present geomagnetic field. Samples with intermediate intensities on the order of 10^{-4} A/m had more scattered directions with a tendency toward westerly declinations and negative inclinations.

Chemical alteration near the surface of the outcrop due to weathering would, if present, manifest itself in a strong difference in the intensities of magnetization of the A and B specimens. In fact both members of a pair were generally found to have almost identical intensities with the B specimen from nearer the surface only slightly weaker. In only two cores were the intensities greatly different, reflecting either the effects of weathering or some other irregularity in composition or history. In general near-surface weathering does not appear to have greatly altered the magnetization of these rocks.

3.2. Pilot study

To provide a preliminary assessment of the magnetization, all of the A specimens were cleaned thermally and all of the B specimens magnetically. Heating was done in a Schonsted furnace in which the residual field in the cooling chamber was 3 nT. To remove any soft components of magnetization acquired while transferring the specimen from the furnace to the magnetometer, the specimens were routinely cleaned in an alternating field of 5 mT after being placed in the field-free region of the magnetometer. In parallel experiments the B specimens were demagnetized in alternating fields (AF) with peak values ranging from 5 to 80 mT, these experiments being done in a field-free region within the superconducting shield of the magnetometer. At each level of AF demagnetization the specimen was kept stationary as it was demagnetized about three orthogonal axes in succession. Both thermal and AF demagnetization were continued until the directions of magnetization became erratic or until the magnetic intensity dropped to 10^{-5} A/m, below which the signal-to-noise ratio of the magnetometer became unacceptable. The main limitation in the demagnetization experiments was imposed not by the sensitivity of the magnetometer but rather by the acquisition of erratic components of magnetization at high levels of demagnetization, this being more pronounced at high temperatures than at high fields. In retrospect these erratic magnetizations appear to have been acquired in transit between the heating and cooling regions of the furnace, the field of the gap having been about 100 nT when these experiments were run.

Three different types of behavior were observed in the 29 pairs of experiments:

(a) *Type 1.* In ten pairs of specimens, an NRM parallel to the present field was totally removed on

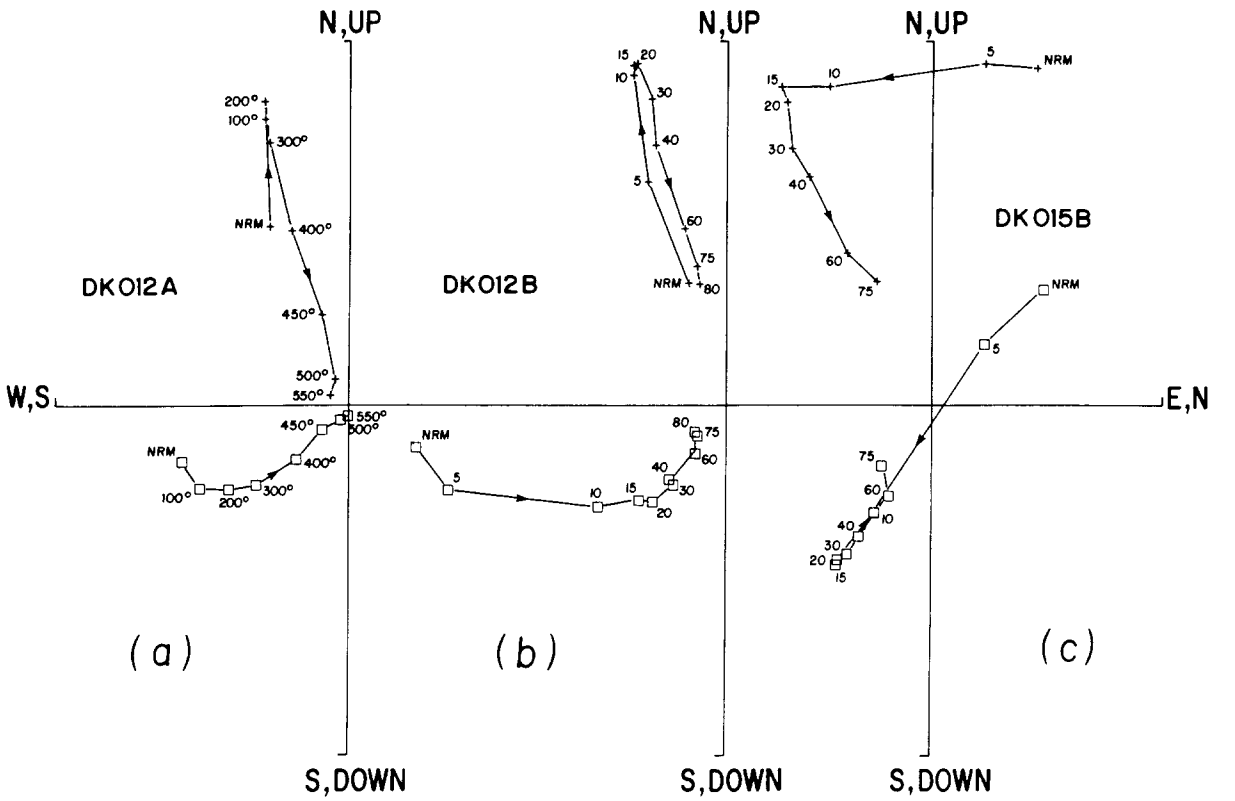


Fig. 5. Partial demagnetization experiments on pilot specimens by heating (a) and in alternating fields (b, c). □ = horizontal components of magnetization; + = components of magnetization in north-south vertical plane. In (b) and (c), peak demagnetizing fields are given in units of mT.

heating the A specimen to 200°C, whereas AF demagnetization up to 80 mT had almost no effect on the adjacent B specimen. The NRM appears to be a low-temperature TRM or a hardened viscous magnetization possessing no useful paleomagnetic information.

(b) *Type 2.* In six pairs of specimens the NRM was totally destroyed in low fields (20 mT) and at low temperatures (200°C or less). No useful stable component of magnetization could be identified.

(c) *Type 3.* In thirteen pairs of specimens a stable component of magnetization was present with blocking temperatures in the range 450–550°C in the A specimens and with coercivities in excess of 80 mT in the B specimens (see specimens DK 012A and DK 012B in Fig. 5). A soft component of magnetization present in most of these specimens was removed at temperatures of 300°C or less and in alternating fields of 20 mT (Fig. 5). For a given pair of specimens, the stable component found after magnetic cleaning invariably had the same magnetic direction as the stable component found from the heating experiments.

The main conclusion to be drawn from the pilot study is that whenever a useful stable component of magnetization is present, it can be found using either thermal or AF cleaning. Since the magnetic directions found by AF cleaning were somewhat less scattered than those found by thermal cleaning, AF demagnetization was used to determine the characteristic direction of the remaining specimens.

The pilot study was used in the following way to design an experimental plan for magnetically

cleaning the rest of the specimens. The thirteen *type 3* pilot specimens had each been demagnetized at nine levels of peak demagnetizing field. These nine magnetic vectors together with the NRM were analyzed using the statistical method of Kirschvink [15], which treats each vector as a point in magnetic vector space and fits these points with line segments each of which describes one component of magnetization. Applying Kirschvink's procedure to different subsets of the complete set of vectors for each specimen revealed that using the three magnetic vectors obtained by demagnetization at levels of 30, 40, and 60 mT yielded the same characteristic direction as using the complete set of ten vectors. Therefore these three demagnetization levels were selected for the analysis of the remaining 61 specimens.

3.3. Results

All the remaining specimens yielded the same three types of results as did the original 29 pilot specimens:

(a) *Type 1.* Eighteen specimens had a very hard component of magnetization parallel to the present field. These were all heated and found to have low blocking temperatures in the range 200–300°C, as seen in DK 032 in Fig. 6. The only exception was a single specimen with a blocking temperature of 550°C. As before, we interpret the single component of magnetization in these specimens as a thermal or hardened viscous overprint parallel to the present field.

(b) *Type 2.* The NRM of twenty four of the specimens was completely unstable, being totally

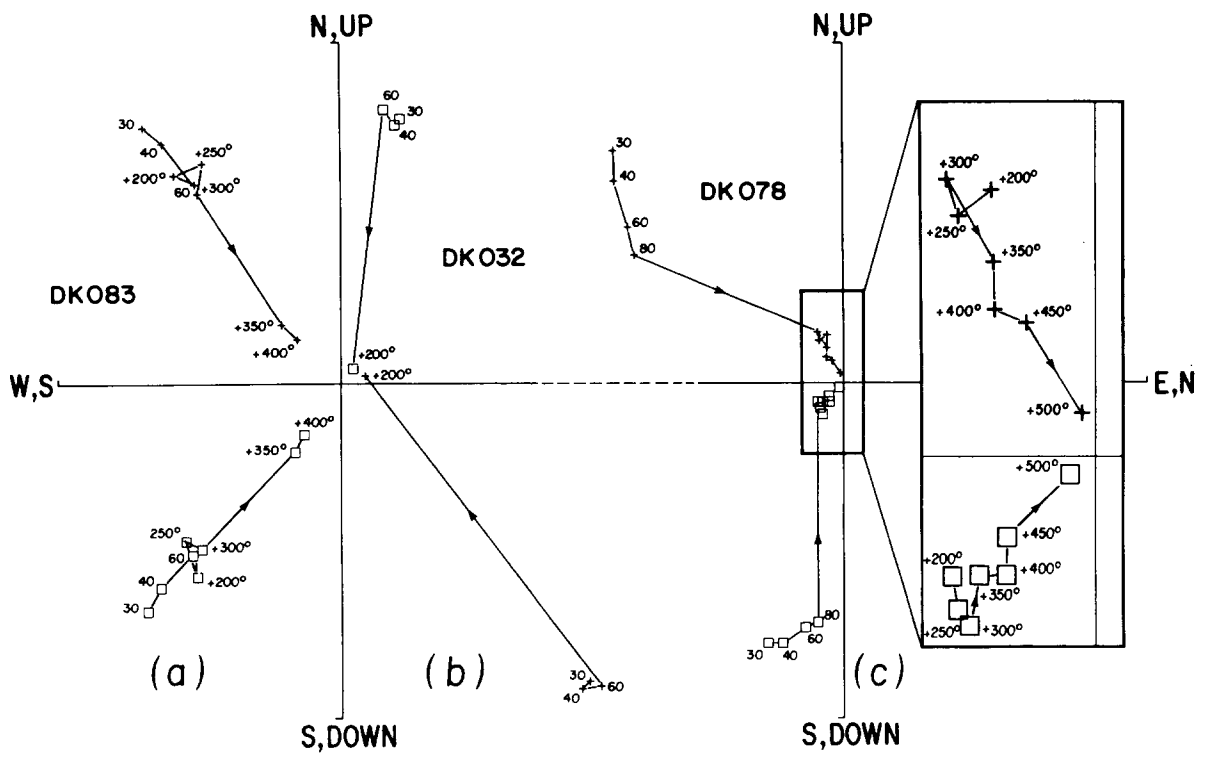


Fig. 6. Demagnetization experiments on specimens with stable (a) and hardened viscous (b) magnetizations. Both types of magnetization are present in (c). Conventions as in Fig. 5.

demagnetized at the first demagnetization step of 30 mT.

(c) *Type 3*. Nineteen specimens possessed stable components of magnetization, the coercivities of which were found to vary considerably. To check whether AF cleaning was yielding the same component of magnetization as thermal would have, specimens with high intensities remaining after AF demagnetization were subsequently thermally demagnetized. As may be seen in specimen DK 083 (Fig. 6), although the thermal results are somewhat noisier for reasons discussed earlier, heating reveals the same stable direction of magnetization as does AF demagnetization.

Magnetizations of type 1 and type 3 were both present in two specimens from the same outcrop (see specimen DK 078 in Fig. 6). AF demagnetization first removed part of the type 3 magnetization. Heating from 200 to 300°C then removed the type 1 magnetization. Finally, heating to 500°C removed the amount of type 3 magnetization that was still present.

3.4. Origin of the stable remanence

Thin sections from five different cores examined in transmitted light revealed that the rocks were originally deposited as a lime mud with carbonate grains about 1 μm in diameter. Subsequent recrystallization produced a limestone with grains 5–10 μm in diameter. The presence of veins of coarser calcite is consistent with the recrystallization having been linked to tectonism (S. Graham, personal communication). Since any depositional magnetization originally present in the lime mud would scarcely have survived the recrystallization, the NRM was acquired either by the mechanical realignment of the original detrital opaque grains during the recrystallization or as a chemical remanence of new opaque minerals that grew during the recrystallization.

To help identify the very fine-grained and highly dispersed magnetic minerals in these limestones five specimens were given isothermal remanence in constant magnetic fields up to 0.7 T (Fig. 7). The results were quite different for different specimens, indicating considerable variation in magnetic mineralogy. Specimen DK 081 with remanence

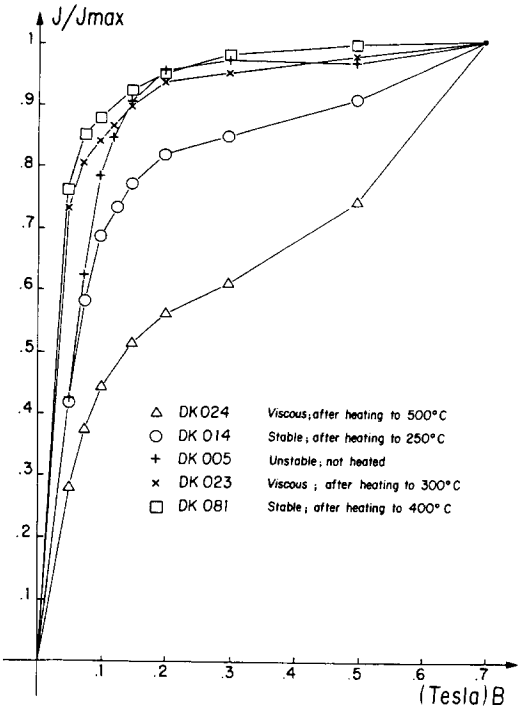


Fig. 7. Isothermal remanence normalized to maximum value as a function of applied constant field B.

type 3 showed near saturation at fields of 0.3–0.4 T, indicating that magnetite is the main magnetic mineral present. Type 1 specimen DK 024 appears to contain both magnetite and hematite. There is no direct correlation between stability and composition. The presence of magnetite is not always an indicator of stability nor is that of hematite always an indicator of instability. However, these experiments show that both magnetite and hematite in varying proportions are present in many of the specimens.

The presence of two distinct magnetic carriers was clearly indicated by the demagnetization experiments described earlier. The carrier of the stable type 3 remanence is characterized by a wide spectrum of coercivities (20 to > 80 mT) and by a wide spectrum of blocking temperatures (200–500°C). This suggests magnetite or titanomagnetite with a broad grain size distribution as a carrier of this magnetization. The carrier of the type 1 magnetization is characterized by high coercivity (80 mT) and low blocking temperatures

TABLE 2

Paleomagnetic results

Site No.	N	D (°)	I (°)	k	α_{95} (°)	VGP (lat., long.)
1	2	235.0	−35.4	10	97.7(?)	38.9°N, 334.9°E
3	4	213.4	−62.0	74	10.7	65.2°N, 310.7°E
4	5	214.3	−57.7	93	8.0	63.6°N, 321.5°E
5	3	218.8	−51.1	25	25.3	57.7°N, 331.2°E
14	1	208.8	−53.3	–	–	66.2°N, 335.2°E
15	3	203.7	−49.7	10	40.6	68.3°N, 348.2°E
17	3	222.1	−44.5	101	12.4	52.4°N, 337.1°E
18	4	237.7	−58.1	40	14.7	46.4°N, 310.5°E
19	3	219.9	−32.9	102	12.2	48.9°N, 349.3°E
20	4	219.7	−44.2	180	6.9	54.1°N, 339.4°E
Mean	10	218.7	−49.4	46	7.2	56.9°N, 332.0°E
						$K = 44$, $A_{95} = 7.4$

(200–300°C). Quite possibly this carrier is fine-grained hematite.

3.5. *Analysis of results*

Stable components of magnetization were found in 32 specimens from ten different sites. These results were analyzed on three different levels. Kirschvink's statistical method was first applied to the set of partial remanent vectors for each specimen to obtain a characteristic direction for that specimen. Next these were combined to obtain site means (Table 2). Finally the site means were combined to yield the mean direction of magnetization for the Kennett Formation: $I = -49°$, $D = 219°$, $A_{95} = 7.2°$. The grouping of site means is quite good ($K = 47$) prior to tectonic correction, as is that of individual specimen directions (Fig. 8). The scatter becomes much greater after tectonic corrections are made. The inescapable conclusion is that the magnetization was acquired after folding of the formation.

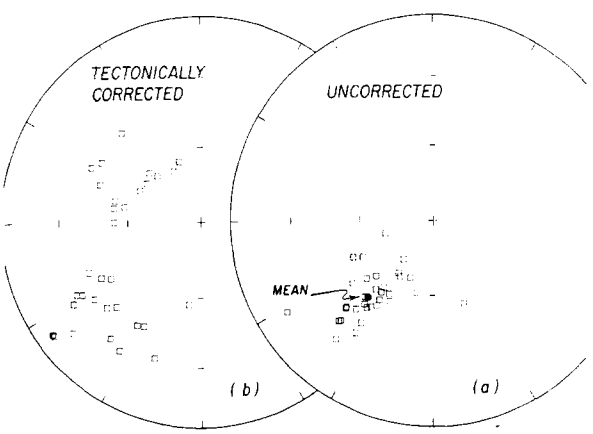


Fig. 8. Magnetic directions for all stably magnetized individual specimens before and after correction for folding within the Kennett Formation. Tectonic corrections were made about horizontal axes parallel to the local strikes of the beds. The age of the magnetization is clearly post-folding.

4. Interpretation

The tectonic significance of these results depends upon the age of the magnetization. Three observations offer clues as to when the magnetization was acquired.

(1) It was acquired after the folding of the Kennett.

(2) It was probably acquired during recrystallization of the limestone that may have been associated with a tectonic or magmatic event.

(3) Since the magnetization of all specimens has a single polarity and since the recrystallization must have occurred over a substantial interval of time, the magnetization was probably acquired during an interval of stable geomagnetic polarity.

Several successive tectonic events are responsible for the folding of the Kennett. The older ones

are recorded in discontinuities in the Redding section whereas the younger are inferred from major regional tectonic events.

A first local stratigraphic discontinuity occurs in the Redding section between the Devonian Kennett and the Mississippian Bragdon Formations. Although unclear, it has been interpreted as an erosional disconformity due to local uplifting of the Kennett [16]. A younger discontinuity occurs between the late Pennsylvanian to early Permian McCloud and the Permian Nosoni Formations. It is partly hidden (or enhanced?) by the diorite intrusion discussed earlier but is regarded as the major disconformity in the section.

Among the regional tectonic events which presumably affected the Eastern Klamath belt, the earliest is the Permo-Triassic Sonoma orogeny; but it does not appear to have been very active this far to the west. A younger and more important event, in this context, is the Jurassic Nevadan orogeny. The latest tectonic phase in the area is observed during the Middle Cretaceous or Early Eocene in the western part of the Klamath Mountains ([17]; see also Roure [8] for a review). Because of the lack of small-scale structural study in the eastern Klamaths, it is difficult to determine the amount of tectonic overprinting in the Redding section due to these orogenies. However, several lines of reasoning suggest that the Nevadan orogeny was, indeed, active within the Eastern Klamath belt. Carey [19] followed by Davis [20] and by Hamilton [5,21] noted the geologic and tectonic similarities between the Blue Mountains, the northern Sierra Nevada and the eastern Klamaths. More recently, Wright and Sharp [22] presented new structural and geochronological data which establish strong correlations between the late Paleozoic to middle Mesozoic tectonic evolution of the whole Klamath Mountains and Sierra Nevada provinces. During the field trip of the French Geological Society in the Western Cordillera of North America, a Nevadan phase of intense tectonic activity was observed in late paleozoic terranes from the northern part of the Sierra Nevada, directly correlative of formations from the Redding section [23]. On the other side, intense isoclinal Jurassic folding is observed in the Foothill belt associated with westward overthrusting of the Western Klamath and Central Metamorphic belts. From all these observations we gather that it is extremely unlikely that the Redding section of the eastern Klamaths was not affected by the Nevadan orogeny. (It might be of interest, here, to note that in the Blue Mountains of eastern Oregon, the Weatherby Formation near the Snake River which has long been considered as homoclinal appeared to be, at closer look, isoclinally folded by the Nevadan orogeny [24].)

The middle Cretaceous to Early Eocene tectonic phase observed in the western part of the Klamath Mountains shows a NNW-SSE direction of the folds, turning NW-SE towards the south, following the arcuate shape of the belts. This folding displays a SW-NE general direction of compression. We observe in Fig. 8a that the directions of magnetization obtained in this paper do not follow a true Fisherian distribution about the mean, but, instead, tend to be aligned along a similar SW-NE direction, suggesting a light tectonic overprint in the folding of the Redding section associated with this early Tertiary tectonic phase. Indeed, if this phase was still gently active this far to the east, Fig. 8 is consistent with a model in which the magnetization observed in the Kennett Formation postdates the Jurassic Nevadan orogeny but predates a later middle Cretaceous to Early Eocene phase.

Younger (Cenozoic) tectonic activity is further displaced towards the west and therefore produces less and less effects in the eastern Klamaths. The structural boundaries between the successive belts of the Klamaths and associated terranes are described as parallel east-dipping thrust faults (see above—second paragraph—and Irwin [4]). It is, then, unlikely that these late phases of tectonism produced much internal deformation in the more eastern belts, but only, eventually, a general tilting of these belts towards the east.

Turning, then, to the two other constraints on the age of the magnetization, we see (Fig. 4) that the recrystallization of the limestone may well have occurred during the time of enhanced groundwater circulation thermally driven by the nearby intrusion [25] of the Shasta Bally batholith at 136 Ma [14]. In this case, the magnetization would have been acquired during the M17 re-

versed interval with an estimated duration of one million years and an estimated age of 134 Ma [26].

The Cretaceous and Tertiary deposits from the Great Valley sequence sit disconformably on top of the eastern Klamaths and the neighbouring terranes. On the other hand, a recent paleomagnetic study [27] indicates that the Klamath Mountains region had achieved its present latitudinal position with respect to the North American craton by Cretaceous time. This rules out the possibility that the magnetization of the Kennett was acquired during a period of normal polarity. Indeed, it would place the eastern Klamaths 12500 ± 1500 km south of its present position with respect to North America, in the early Lower Cretaceous.

Our interpretation is based on the assumption that the recrystallization and magnetization of the limestone took place during the cooling of the Shasta Bally batholith at 136 Ma [14]. The paleofield for the Kennett is shown as *KL* in Fig. 9 together with the reversed paleofield direction that would be expected 140 Ma before present if the Kennett had undergone no translation or rotation relative to the stable craton. The same information is displayed using paleopoles in Fig. 10. The value of the paleomagnetic declination of the Kennett Formation is $70 \pm 11°$ greater than the expected value, corresponding to a clockwise rotation by this amount of the Eastern Klamath terrane relative to the stable craton. Part of this rotation probably occurred during the latter part of the Tertiary. Magill and Cox [28] report that the Oligocene western Cascade Range to the east and north of the Klamaths has undergone a clockwise rotation of $27 \pm 7°$ during the past 25 millions years. Since the older western Cascade volcanics rest on Klamath basement at the south end of the range, they propose that the Klamath Mountains also underwent 27° of post-Oligocene rotation. The same result has been proposed, more recently by Craig et al. [29], who, however, obtain a smaller value for this rotation. Removing this younger rotation yields the vector *KL'* in Fig. 9, corresponding to about 43° of pre-Miocene rotation of the Eastern Klamath terrane. The paleomagnetic inclination of the Kennett indicates a colatitude of 60° whereas the distance from the site to the paleomagnetic pole for the stable craton at 140 Ma

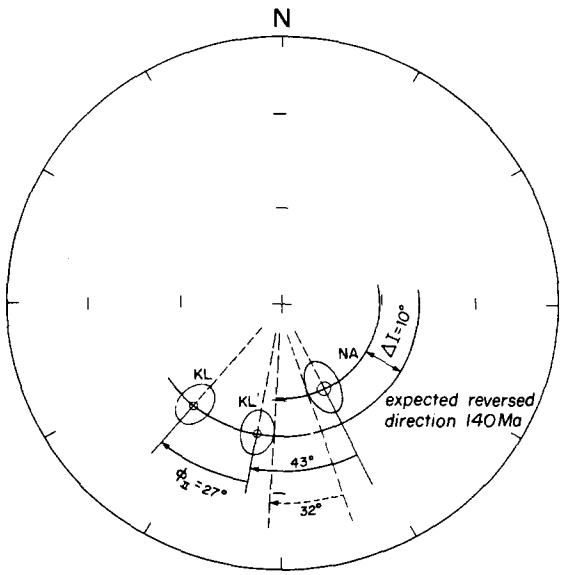


Fig. 9. Paleomagnetic direction for the Kennett (*KL*) and that expected at the sampling site (*NA*) if the age of the magnetization is 140 Ma. The latter is based on global mean pole of Harrison and Lindh [30]. *KL'* is the paleomagnetic direction of the Kennett Formation after removing the effects of 27° of presumed post-Oligocene clockwise rotation.

is 50° [30] the corresponding amount of poleward displacement being $10 \pm 9°$, which is hardly significant (Table 3). When compared with the paleomagnetic pole for the stable craton at 130 Ma, the results show a significant poleward displacement of $13 \pm 9°$, which corresponds to a northward translation, roughly parallel to the western margin of North America (Fig. 10), of at least 400 km. The pole for North America at 130 Ma has been obtained using an age window of 30 Ma for which the best weighted median age is computed at 135 Ma (see Harrison and Lindh [30]). Although the authors conclude that the discrepancies between the mean age and the middle of the age window are not significant, this suggests that the 130-Ma pole describes best the position of North America at the time of magnetization of the Kennett Formation. Under our assumptions, the paleomagnetism of the Kennett is consistent with a nearly autochthonous eastern Klamath terrane or with one that is allochthonous but came very close to its present location relative to North America prior to 140 Ma before present. The

TABLE 3

Rotations and displacements for different assumed ages of magnetization

Age (Ma)	Pole for North America		Declination $D_{exp} \pm \Delta D_{exp}$ (°)	Rotation $R \pm \Delta R$ (°)	Colatitude P_0 (°)	Displacement $d \pm \Delta d$ (°)
	lat., long.	A_{95} (°)				
130	66.2°N, 164.3°E	5.8	147.9 ± 8.0	70.8 ± 11.7	46.8	12.9 ± 9.4
140	66.0°N, 156.5°E	5.6	148.3 ± 7.3	70.4 ± 11.3	49.9	9.8 ± 9.3
150	66.6°N, 147.8°E	5.7	150.2 ± 7.1	68.5 ± 11.2	53.1	5.9 ± 9.3

Poles for 130–150 Ma are from Harrison and Lindh [30].

P_0 is the distance from the site (40.78°N, 237.58°E) to the pole of this age.

$\Delta D_{exp} = \sin^{-1}(\sin A_{95}/\sin P_0)$.

R, the amount of clockwise rotation, is given by $R = D_{obs} - D_{exp}$, where $D_{obs} = 218.7°$.

ΔR, the 95% confidence limit for R, is given by $\Delta R = (\Delta D_{obs}^2 + \Delta D_{exp}^2)^{1/2}$.

ΔD_{obs}, the error in D_{obs}, is given by $\Delta D_{obs} = \sin^{-1}(\sin 7.4°/\sin 59°) = 8.6°$, where 59° is the distance from the site to the mean paleomagnetic pole for this site.

d is the displacement of the terrane toward the paleomagnetic pole for North America. Negative values indicate displacement in a southeasterly direction.

Δd, the error in the displacement, is given by $\Delta d = (A_{95}^2 + 7.4^2)^{1/2}$, where 7.4° is the error in the Kennett pole.

conclusions drawn are only slightly sensitive to small variations in the age of the magnetization. Assumed ages of 130, 140, and 150 Ma (Table 3) give rotations of 71°, 70°, and 68° and similarly the amount of poleward displacement indicated by the paleomagnetic data is just above, at or below the experimental uncertainties.

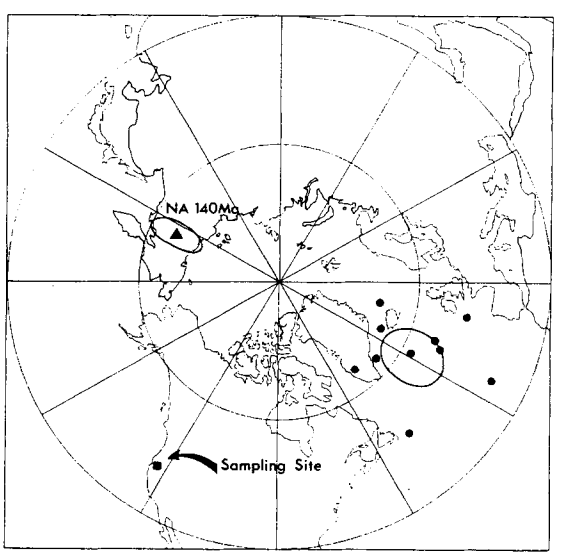


Fig. 10. Paleomagnetic poles for individual sites (solid circles) and 95% confidence circle about mean pole for Kennett Formation. Paleomagnetic pole for the stable North American craton is shown for the age 140 Ma before present (triangle).

An alternative explanation for the discrepancy between the paleomagnetic direction of the Kennett Formation and the expected North American direction would be the existence of 40° of tilting towards the west about an almost north-south horizontal axis (parallel to the trend of the belts) posterior to the magnetization. However, we have little confidence in this interpretation since the polarity of the tectonic tilt involved is in contradiction with the observed general vergence of all the belts of this part of the western Cordillera [4]. Our main conclusions would, on the other hand, hardly be affected by the occurrence of a Cenozoic eastward tilting of the block, as may be supported by the Geology. Indeed if this were the case, the true paleofield direction of the eastern Klamaths would be displaced towards the west (larger declination and a shallower inclination) and it would imply a slightly larger clockwise rotation with, in addition, a more significant northward translation.

5. Discussion

The firmest conclusion that can be drawn from the paleomagnetic results is that the Eastern

Klamath Mountains terrane has undergone 70 ± 11° of post-Jurassic clockwise rotation, of which at least 43° is pre-Miocene. This conclusion remains valid if the magnetization was acquired at any time between 150 Ma and present, although the amount of rotation is slightly changed. The tectonic implications of this clockwise rotation can best be considered in the context of regional tectonic patterns.

The second conclusion is that the Eastern Klamath terrane may have undergone a few hundred kilometers of northward displacement along the western margin of North America since the end of the Jurassic.

The dominant tectonic pattern in the Klamath Mountains is that of a set of arcuate belts concave to the east, as if the belts had been wrapped around a central buttress. If, in fact, the tectonic belts of the Klamath had been deformed in this way, the belts in the south would be rotated in a counterclockwise sense, those on the west would be unrotated, and those to the north would be rotated in a clockwise sense. The discovery of clockwise rather than the counterclockwise rotation predicted by this simplistic model indicates that a simple wrap-around tectonic model does not apply to Klamath tectonics.

Turning to broader regional tectonic patterns, Carey [19] and more explicitly Hamilton [5] saw the Blue Mountains of eastern Oregon, the Klamaths, and the Paleozoic belt of the northern Sierra Nevada as three tectonic blocks that had once been adjacent to each other along a once continuous belt of rocks west of an ancestral Sierra Nevada batholith. Hamilton [5] proposed that during a single rifting event the Blue Mountains underwent a large clockwise rotation as the Klamaths moved westward with perhaps a slight counterclockwise rotation, whereas the southern block stayed in place next to the Sierra Nevada batholith. Dickinson and Thayer [31] also proposed on geologic grounds that the Blue Mountains had undergone clockwise rotation during an episode of rifting, but in their model the rifting was a local one in which the Blue Mountains underwent a rotation of approximately 60° about a local Euler pole located immediately north of the range. From a paleomagnetic study of the Blue Mountains, Wilson and Cox [32] found that this terrane had, in fact, rotated 60° as predicted by Hamilton [5] and Dickinson and Thayer [31]. The new paleomagnetic results from the eastern Klamath terrane are not consistent with Hamilton's model if a single rifting event occurred in the Tertiary, as he proposed. However, if rifting of the Klamaths with little or no rotation occurred prior to the Early Cretaceous as is favored by local geology [33], then quite possibly the observed rotation is the result of a later independent event.

A clockwise rotation of about 70° has been observed paleomagnetically immediately north of the Klamath Mountains in the Eocene rocks of the Oregon Coast Range. This rotation has been accounted for by a model in which two subduction zones were active simultaneously during the Eocene, permitting the crust between them to rotate in a clockwise sense [28,34]. Magill et al. [35] estimate that 46° of rotation occurred by this mechanism during the Oligocene. The similarity of magnitudes of the rotations in the Oregon Coast Range, in the Klamaths and in the Blue Mountains opens the possibility that the three blocks underwent a Tertiary rotation as part of the same crustal block.

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